NAVRACHANA UNIVERSITY

Teaching Plan

Core Java Programming (CS348)

Credit : 2

Faculty	<table><tr><td>Name</td><td>Contact</td><td>Office Hours</td><td>Office Location</td></tr><tr><td>Vivek Bhatt</td><td>vivek.bhatt@nuv.ac.in</td><td></td><td></td></tr></table>
School	SET
Semester	Spring (Jan to June 2025)
Lecture time & Weedays	<table><tr><td>SECTION_1</td><td>Monday</td><td>10:00:00-11:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr><tr><td></td><td>Tuesday</td><td>10:00:00-11:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr><tr><td>SECTION_2</td><td>Tuesday</td><td>12:00:00-13:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr><tr><td></td><td>Wednesday</td><td>12:00:00-13:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr></table>
Pre-requisite	
Co-requisite	
Anti-requisite	
Course Description	This course of study builds on the skills gained by students in Java Fundamentals or Java Foundations to help advance Java programming skills. Students will design object-oriented applications with Java and will create Java programs using hands-on, engaging activities.
Course Abstract	Through a combination of theoretical understanding and practical exercises, the course equips students with the skills to design efficient, scalable, and dynamic Java-based applications, preparing them for both academic and professional challenges in software development.
Course Objectives	1. Introduce Java as a programming language. 2. Introduce Java as an Object-Oriented programming language. 3. Introducing the Concepts of Exception Handling and Multithreading 4. Introducing the Concepts of Java Collections Framework and Database Programming 5. Introducing the concepts of Swing GUI and Servlets.
Course Outcomes	CO1. Learn the Java programming language: its syntax, idioms, patterns, and styles CO2.The skills to apply OOP concepts in Java programming in problem solving. CO3. Understand the concepts of Exception Handling and Multithreading CO4. Learn the essentials of the Java Collections Framework and Database Programming CO5. Learn Swing GUI and Servlets
Course Typology	Theory
Course Outline	Unit - 1 : Java Basics – History of Java, Java Features, comments, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expressions, type conversion and casting, enumerated types, control flow-block scope, conditional statements, loops, break and continue statements, simple java program, arrays, input and output, formatting output, Review of OOP concepts, constructors, methods, static fields and methods, access control, this reference, overloading methods and constructors, recursion, garbage collection, building strings, exploring string class, Enumerations, Generics Unit - 2 : Object Oriented Concepts – Inheritance concept, benefits of inheritance, Super classes and Sub classes, Member access rules, Inheritance hierarchies, super uses, preventing inheritance: final classes and methods, casting, polymorphism-dynamic binding, method overriding, abstract classes and methods, the Object class and its methods. Unit - 3 : Exception handling and Multithreading– Dealing with errors, benefits of exception handling, the classification of exceptions-exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, re-throwing exceptions, exception specification, built in exceptions, creating own exception sub classes, Multithreading – Differences between multiple processes and multiple threads, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, interthread communication, thread groups, daemon thread Unit - 4 : Java Collections Framework and Database Programming – Introduction to Java Collections, Overview of Java Collection frame work, Commonly used Collection classes – Array List, Linked List, HashSet, HashMap, TreeMap, Collection Interfaces –, Using javautil Files – streams- byte streams, character streams, text Input/output, binary input/output, random access file operations, File management using File class, Using javaio., Database Programming using RDBMS (MySQL), Developing CRUD applications with CLI, GUI and Web Applications. Unit - 5 : Swing GUI, Servlets and JSP Package Organizing Classes and Interfaces in Packages , Package as Access Protection , Defining Package ,CLASSPATH Setting for Packages , Making JAR Files for Library Packages Import and Static Import Naming Convention For Packages. Event-Driven Programming in Java, Event- Handling Process, Event Handling Mechanism, The Delegation Model of Event Handling, Event Classes, Event Sources, Event Listeners, Adapter Classes as Helper Classes in Event Handling. Desktop Applications with Swing/AWT. Web Application Development using Java, Java Servlet, Java Server Pages, MVC Development, Introduction to Spring Framework.
Pedagogy	Flipped classrooms, Presentation, Real-time problem solving with Java programms
Expectations from Students	Laptops with JDK (at least 1.8) installed
Assessment / Evaluation	
Project / Assignment Details	Continuous Evaluation (60%) 1. Assignment (OOPS concepts + Multi-threading) (20%) 2. Assignment (DBMS) (20%) 3. Assignment (Web Applications) (20%) Final Evaluation (40%) Combined project with DBMS course
Course Material	1. The complete reference, 7th edition, Herbert Schildt, TMH. 2. Java for Programmers, PJDeitel and HMDeitel, Pearson education / Java: How to Program PJDeitel and HMDeitel,8th edition, PHI 3. Core Java, Volume 1- Fundamentals, eighth edition, Cay Horstmann and Gary Cornell, Pearson education.
Additional Information	This course requires serious attention to learn a programming language from scratch

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
Java Basics	1	History of Java, Java Features, comments, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expressions			
Java Basics	2	Type conversion and casting, enumerated types, control flow-block scope			
Java Basics	3	Conditional statements, loops, break and continue statements, simple java program, arrays, input and output, formatting output			
Java Basics	4	Review of OOP concepts			
Java Basics	5	Constructors, methods, static fields and methods, access control, this reference, overloading methods and constructors, recursion			
Java Basics	6	Garbage collection, building strings, exploring string class, Enumerations, Generics			
Object Oriented Concepts	7	Inheritance concept, benefits of inheritance			
Object Oriented Concepts	8	Super classes and Sub classes, Member access rules, Inheritance hierarchies			
Object Oriented Concepts	9	Super uses, preventing inheritance: final classes and methods			
Object Oriented Concepts	10	Casting, polymorphism-dynamic binding			
Object Oriented Concepts	11	Method overriding, abstract classes and methods			
Object Oriented Concepts	12	The Object class and its methods.			
Exception Handling and Multithreading	13	Dealing with errors, benefits of exception handling			
Exception Handling and Multithreading	14	The classification of exceptions, Exception hierarchy			
Exception Handling and Multithreading	15	Checked exceptions and unchecked exceptions			
Exception Handling and Multithreading	16	Usage of try, catch, throw, throws and finally			
Exception Handling and Multithreading	17	Rethrowing exceptions, exception specification, built in exceptions, creating own exception sub classes			
Exception Handling and Multithreading	18	Multithreading – Differences between multiple processes and multiple threads, thread stat			
Exception Handling and Multithreading	19	Creating threads, interrupting threads, thread priorities, synchronizing threads, interthread communication, thread groups, daemon thread			
Java Collections Framework and Database Programming	20	Introduction to Java Collections, Overview of Java Collection frame work, Commonly used Collection classes – Array List, Linked List, HashSet, HashMap, TreeMap, Collection Interfac			
Java Collections Framework and Database Programming	21	Using javautil Files – streams- byte streams, character streams, text Input/output, binary input/output, random access file operation			
Java Collections Framework and Database Programming	22	File management using File class, Using javaio			
Java Collections Framework and Database Programming	23	Database Programming using RDBMS (MySQL), Developing CRUD applications with CLI, GUI and Web Applications			
Swing GUI, Servlets and JSP	24	Package Organizing Classes and Interfaces in Packages , Package as Access Protection			
Swing GUI, Servlets and JSP	25	Defining Package ,CLASSPATH Setting for Packages			
Swing GUI, Servlets and JSP	26	Making JAR Files for Library Packages Import and Static Import Naming Convention For Packages			
Swing GUI, Servlets and JSP	27	Event-Driven Programming in Java, Event- Handling Process, Event Handling Mechanism			
Swing GUI, Servlets and JSP	28	The Delegation Model of Event Handling, Event Classes, Event Sources, Event Listeners			
Swing GUI, Servlets and JSP	29	Adapter Classes as Helper Classes in Event Handling. Desktop Applications with Swing/AWT			
Swing GUI, Servlets and JSP	30	Web Application Development using Java, Java Servlet, Java Server Pages, MVC Development, Introduction to Spring Framework			

Publish Status	Published